

GuardShield AI: Complete Project & Research Guide

B.Tech CSE (AIML) Final Year Minor/Major Project & 7th Sem IEEE Research Paper Plan

Document Purpose: This comprehensive reference guide covers both the **Software Implementation Plan (Minor & Major Project)** and the **Academic Research Paper Roadmap (7th Sem IEEE/Springer Publication)** for *GuardShield AI* — a real-time sidecar proxy for LLM prompt injection defense and hallucination mitigation.

PART 1: SOFTWARE PROJECT BLUEPRINT (MINOR & MAJOR)

1.1 What We Are Building & Deployment Context

At its core, **GuardShield AI** is a **Backend Security Middleware (API Proxy)** paired with an **Interactive Web App & Telemetry Dashboard**. It acts as a digital security guard sitting between user applications and Large Language Models (LLMs).

- **Enterprise Deployment:** Sits as a microservice between corporate apps (e.g., Banking/Healthcare Chatbots) and LLM APIs (OpenAI/Claude/Ollama), filtering attacks before the LLM processes them.
- **User / Browser Deployment:** Functions as a Chrome Extension overlaying real-time factuality confidence badges directly on `chatgpt.com` or `claude.ai` interfaces.

1.2 System Data Flow Architecture

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[User Prompt] ████ [GuardShield Pre-Scan: DeBERTa Injection Filter] ████ (Passed) ████ [Local LLM: Ollama Llama-3.1-8B] ████ (Streaming Tokens) ████ [Web App / UI] ████ [Clean Response / Flagged Alert] ████ [GuardShield Post-Scan: Logit Entropy & NLI Engine] ████
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1.3 Semester Deliverables Breakdown

Semester	Core Software Deliverables	Expected Milestone
7th Sem (Minor)	• DeBERTa-v3 Prompt Injection & Jailbreak Classifier • Token Logit Entropy & NLI Hallucination Scoring Engine • Benchmark evaluation on HaluEval & JailbreakBench	Working Model Pipeline & Draft Research Paper
8th Sem (Major)	• Async FastAPI SSE Streaming Sidecar Proxy • Autonomous LLM Red-Teaming Adversarial Attack Agent • Live Security Telemetry Dashboard (React / Streamlit) • (Bonus) iOS App / Chrome Extension Demo	Production Proxy + Live UI + Published Paper

1.4 Tech Stack & MacBook M4 Pro Zero-Cost Setup

- **Local Inference:** Ollama running llama3.1:8b / qwen2.5:14b natively via Metal (MPS) acceleration.
- **Classifiers:** Hugging Face Transformers (`DeBERTa-v3-small`), PyTorch, ONNX Runtime.
- **Proxy Backend:** Python FastAPI (Async SSE streaming) / Docker.
- **Dashboard:** Streamlit / React + TailwindCSS.

PART 2: 7TH SEMESTER IEEE/SPRINGER RESEARCH PAPER PLAN

2.1 Evolution of AI Guardrails (Past, Present & Future)

Era	Dominant Paradigm	Key Limitations & Research Gaps
Past (2022-23)	Static Regex blocklists & heavy GPT-4 AI-Judges	Added 1-3s latency, expensive, easily bypassed via base64 or rephrasing
Present (2024-26)	Small SLM Guardrails (Llama-Guard, DeBERTa) & RAG NLI Checks	Post-hoc verification causes streaming delays; judges vulnerable to stealth attacks
Future (2026+)	Real-Time Streaming Logit Inspection & Agentic Safety	Need sub-15ms early token-termination algorithms during token generation

2.2 Selected Research Gap for Your Paper

Target Research Gap: Early-Token Termination for Streaming Hallucination Mitigation

Problem: Existing NLI fact-checkers wait for full sentence generation before evaluating factuality, ruining real-time streaming user experience.

Novel Solution: GuardShield introduces early-token logit entropy monitoring during streaming. When token probability uncertainty spikes, windowed NLI checks trigger immediately to correct or terminate the stream **mid-sentence** under 25ms overhead.

2.3 Research Paper Metadata (Ready for Submission)

Proposed Title: *GuardShield: Low-Latency Streaming Hallucination Detection via Token Entropy Thresholding and Windowed NLI Entailment*

Target Publication Venues: IEEE International Conference Student Track / Springer LNCS / Scopus-indexed Conference.

Keywords: Trustworthy AI, Hallucination Mitigation, Prompt Injection Defense, Real-Time Guardrails, Natural Language Inference.

2.4 Paper Section Outline & Benchmarks

- **Section I: Introduction** — The rise of LLM applications, risks of hallucinations and prompt injections, research motivation.
- **Section II: Related Work** — Review of Llama-Guard, NeMo Guardrails, SelfCheckGPT, and their latency limitations.
- **Section III: Proposed GuardShield Framework** — Mathematical formulation of Token Logit Entropy and NLI cross-encoder verification.
- **Section IV: Experimental Setup** — Evaluation on *HaluEval*, *HALC-Bench*, and *JailbreakBench* datasets using PyTorch MPS.
- **Section V: Results & Comparative Analysis** — Accuracy vs. Latency Trade-Off graphs, F1-scores, and ROC-AUC curves.
- **Section VI: Conclusion & Future Work** — Extension into autonomous agentic tool-call security.

Action Plan: Use Part 1 for your college project synopsis and Part 2 as your 7th semester IEEE research paper outline.